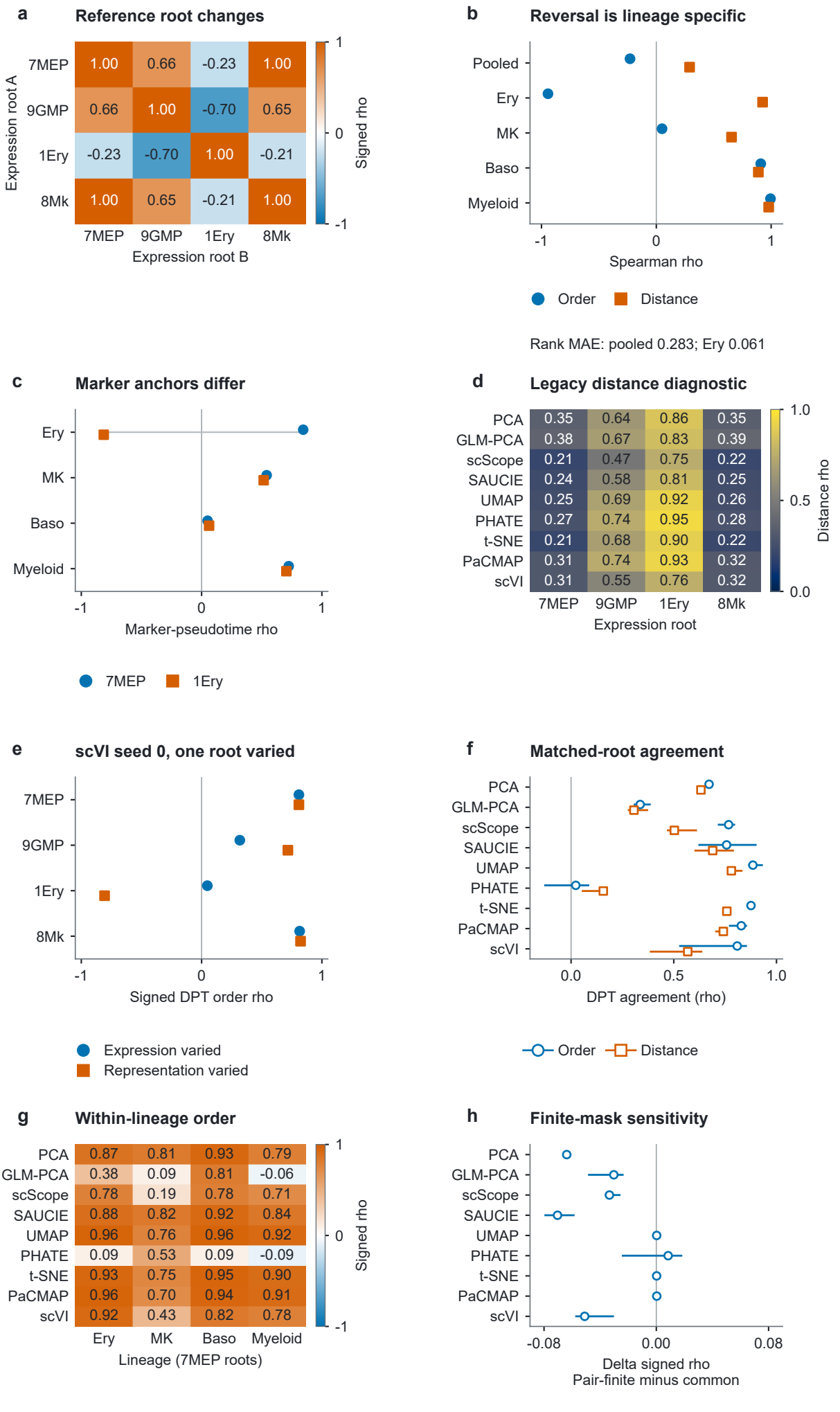

a
Reference root changes
7MEP
9GMP
1Ery
8Mk
1.00 0.66 -0.23 1.00
0.66 1.00 -0.70 0.65
-0.23 -0.70 1.00 -0.21
1.00 0.65 -0.21 1.00
Expression root A
Expression root B
Signed rho
1
0
-1
b
Reversal is lineage specific
Pooled
Ery
MK
Baso
Myeloid
-1
0
1
Spearman rho
Order
Distance
Rank MAE: pooled 0.283; Ery 0.061
c
Marker anchors differ
Ery
MK
Baso
Myeloid
-1
0
1
Marker-pseudotime rho
7MEP
1Ery
d
Legacy distance diagnostic
PCA 0.35 0.64 0.86 0.35
GLM-PCA 0.38 0.67 0.83 0.39
scScope 0.21 0.47 0.75 0.22
SAUCIE 0.24 0.58 0.81 0.25
UMAP 0.25 0.69 0.92 0.26
PHATE 0.27 0.74 0.95 0.28
t-SNE 0.21 0.68 0.90 0.22
PaCMAP 0.31 0.74 0.93 0.32
scVI 0.31 0.55 0.76 0.32
7MEP
9GMP
1Ery
8Mk
Expression root
Distance rho
1.0
0.5
0.0
e
scVI seed 0, one root varied
7MEP
9GMP
1Ery
8Mk
-1
0
1
Signed DPT order rho
Expression varied
Representation varied
f
Matched-root agreement
PCA
GLM-PCA
scScope
SAUCIE
UMAP
PHATE
t-SNE
PaCMAP
scVI
0.0
0.5
1.0
DPT agreement (rho)
Order
Distance
g
Within-lineage order
PCA 0.87 0.81 0.93 0.79
GLM-PCA 0.38 0.09 0.81 -0.06
scScope 0.78 0.19 0.78 0.71
SAUCIE 0.88 0.82 0.92 0.84
UMAP 0.96 0.76 0.96 0.92
PHATE 0.09 0.53 0.09 -0.09
t-SNE 0.93 0.75 0.95 0.90
PaCMAP 0.96 0.70 0.94 0.91
scVI 0.92 0.43 0.82 0.78
Ery
MK
Baso
Myeloid
Lineage (7MEP roots)
Signed rho
1
0
-1
h
Finite-mask sensitivity
PCA
GLM-PCA
scScope
SAUCIE
UMAP
PHATE
t-SNE
PaCMAP
scVI
-0.08
0.00
0.08
Delta signed rho
Pair-finite minus common